Temperature Sensor API Architecture Documentation

Feefty Kata Project

Abdelali ICHOU

Harvest - Feefty

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Sommaire

Temperature Sensor API Architecture Documentation

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Start The Application

Start the db

```
docker compose up -d postgres
```

Build & Start the app

```
npm run start
```

Coverage report inside root/converage folder

```
npm run test:cov
```

---

1. Framework Choice

Why NestJS over Express?

I initially considered **Express.js** for its simplicity. However, after evaluating the requirements (hexagonal architecture, CQRS, dependency injection, contract-first), i've gone with **NestJS** for the following reasons:

CRITERIA	NESTJS
Dependency Injection	Built-in, module-based
CQRS / Mediator	`@nestjs/cqrs` built-in instead of manual Engine primary/secondary ports implementation
Module system	First-class modules with imports/exports
TypeScript	Native by default
Testing	`Test.createTestingModule()` with overrides
OpenAPI	`@nestjs/swagger` (supports static YAML serving)

The decisive factor: NestJS provides a **built-in CQRS module** (`@nestjs/cqrs`) with `CommandBus`, `QueryBus`, `@CommandHandler`, `@QueryHandler`, which mean i don't need to create a Mediator pattern Engine that links the commands/queries by their domain validators/usecases from scratch as we do in Java.

> *Sources: - Node.js Frameworks comparison - Express vs NestJS - Command Bus Design Pattern - NestJS Project Structure Best Practices - Hexagonal Architecture in NestJS*

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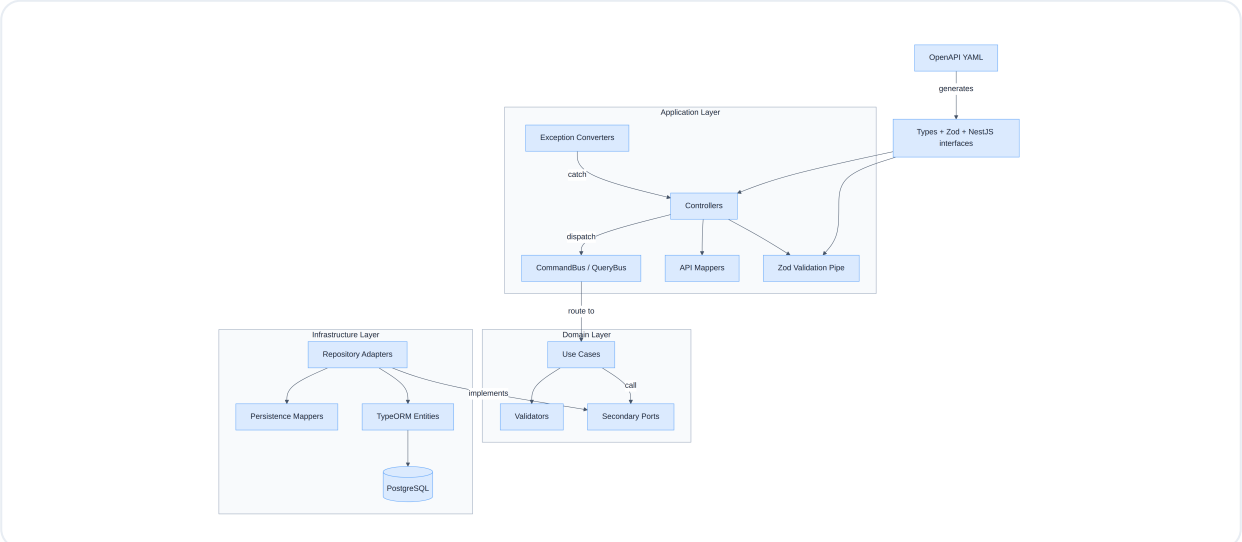
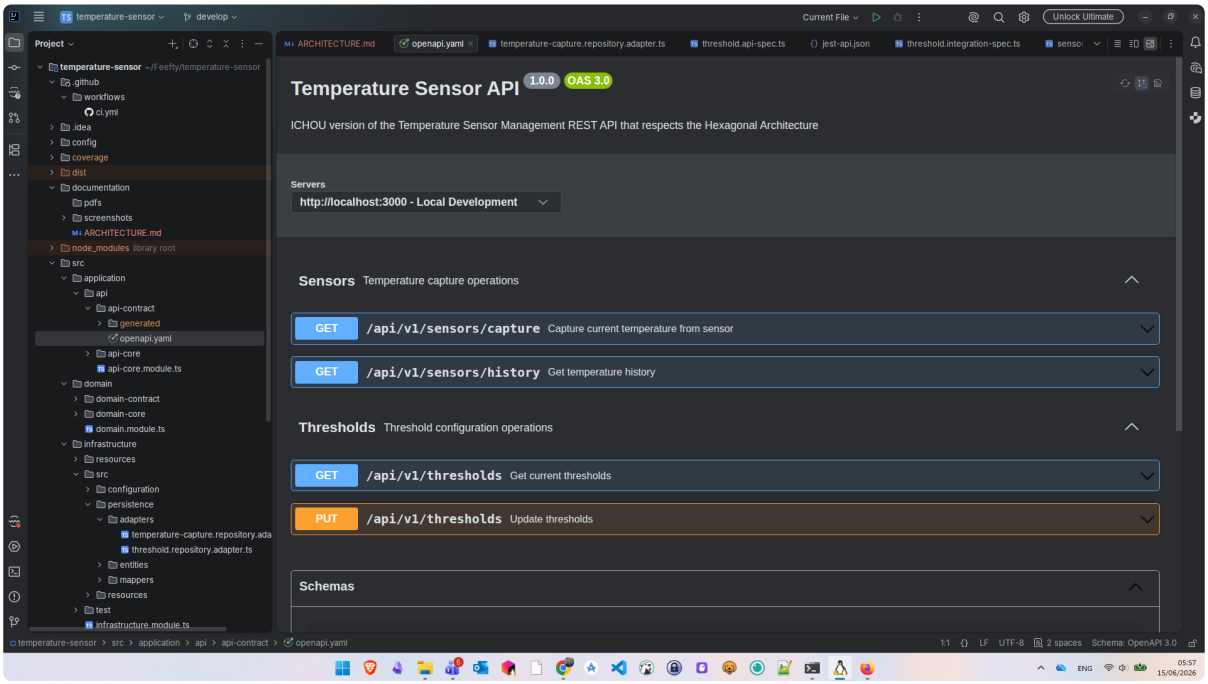
2. Architecture Overview

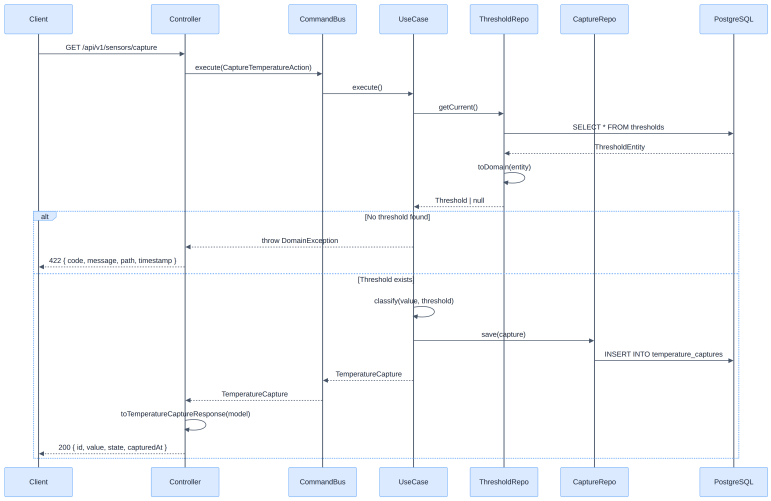


Swagger documentation

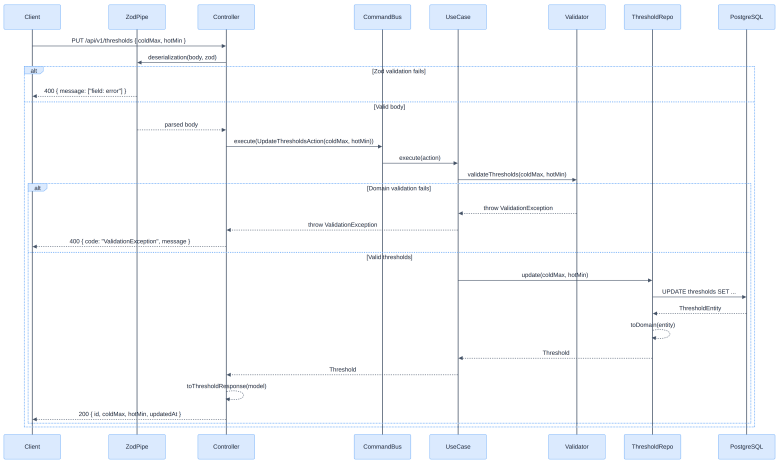


Endpoints Swagger Doc

Capture Temperature Endpoint Sequence Diagram



Update Thresholds Sequence Diagram



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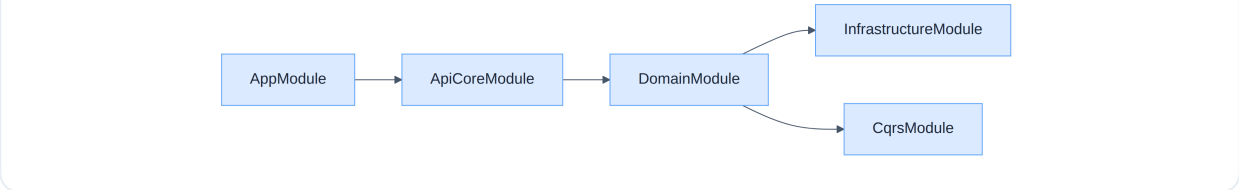
3. Project Structure

Each module follows a `src/` + `test/` convention (inspired by Java's `src/main/` + `src/test/`):

```
temperature-sensor/
├── src/
│   ├── application/
│   │   └── api/
│   │       ├── api-contract/
│   │       │   ├── openapi.yaml              ← THE source of truth
│   │       │   └── generated/                ← auto-generated (types, zod, nestjs)
│   │       ├── api-core/
│   │       │   ├── src/
│   │       │   │   ├── controllers/          ← implements generated interfaces
│   │       │   │   ├── error.converter/      ← DomainException → HTTP response
│   │       │   │   ├── interceptors/         ← logging
│   │       │   │   ├── mappers/              ← domain model → API response mappers
│   │       │   │   └── configuration/        ← Zod desrialization
│   │       │   └── test/
│   │       │       ├── api/                  ← test api error responses, mocking domain
│   │       │       ├── exception/            ← test exceptions converter response
│   │       │       └── integration/          ← full stack + testcontainers
│   │       └── api-core.module.ts
│   │
│   ├── domain/
│   │   ├── domain-contract/
│   │   │   ├── command/
│   │   │   │   ├── action/                ← CaptureTemperatureAction, UpdateThresholdsAction
│   │   │   │   └── query/                 ← GetThresholdsQuery, GetTemperatureHistoryQuery
│   │   │   ├── exceptions/                ← DomainException, ValidationException
│   │   │   ├── models/                    ← TemperatureCapture, Threshold, TemperatureState
│   │   │   └── ports/secondary/           ← repository interfaces ( if no built in @nestjs/cqrs →,
we add primary ports )
│   │   ├── domain-core/
│   │   │   ├── src/
│   │   │   │   ├── usecase/               ← CaptureTemperature, GetHistory, GetThresholds,
UpdateThresholds
│   │   │   │   └── validation/            ← threshold business rules validator
│   │   │   └── test/
│   │   │       ├── usecase/               ← jest.fn() mocks on ports
│   │   │       └── validation/            ← pure unit tests
│   │   └── domain.module.ts
│   │
│   ├── infrastructure/
│   │   ├── src/
│   │   │   ├── configuration/             ← app.config, database.config
│   │   │   ├── persistence/
│   │   │   │   ├── adapters/              ← implements secondary ports
│   │   │   │   ├── entities/              ← TypeORM entities
│   │   │   │   └── mappers/               ← entity ↔ domain model mappers
│   │   │   └── resources/db/
│   │   │       ├── migrations/            ← SQL schema
│   │   │       └── seeds/                 ← default data
│   │   ├── test/
│   │   │   ├── persistence/               ← testcontainers adapter tests
│   │   │   └── setup/                     ← test database utility
│   │   └── infrastructure.module.ts
│   │
│   ├── shared/dinjection/tokens/          ← DI injection tokens (Symbols)
│   ├── app.module.ts                      ← root module
│   └── main.ts                            ← bootstrap
│
├── config/                                ← .env, .env.test, .env.example
```

```
├── Dockerfile                          ← multi-stage build
├── docker-compose.yml                  ← app + PostgreSQL
├── openapi-ts.config.ts                ← hey-api generator config
├── jest.config.ts                      ← unit test config
└── .github/workflows/ci.yml            ← GitHub Actions pipeline
```

Module Dependency Chain



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4. Contract-First Approach

Philosophy

The OpenAPI specification (`openapi.yaml`) is the **single source of truth**. We never write DTOs, request types, or response types manually. Everything is generated.

Generation Pipeline

```
openapi.yaml → @hey-api/openapi-ts → generated/
                                       ├── types.gen.ts    (TypeScript types + runtime enums)
                                       ├── zod.gen.ts      (Zod schemas for runtime validation)
                                       └── nestjs.gen.ts   (Controller interfaces)
```

Configuration (`openapi-ts.config.ts`):

```
import { defineConfig } from '@hey-api/openapi-ts';

export default defineConfig({
  input: './src/application/api/api-contract/openapi.yaml',
  output: { path: './src/application/api/api-contract/generated' },
  plugins: [
    { name: '@hey-api/typescript', enums: 'javascript' },
    'zod',
    'nestjs',
  ],
});
```

What Gets Generated

SOURCE (YAML)	GENERATED	USAGE
`schemas.TemperatureState` enum	`const TemperatureState` (runtime object)	Used in tests for assertion without hardcoding
`schemas.UpdateThresholdsRequest`	`zUpdateThresholdsRequest` (Zod schema)	Used by `ZodValidationPipe` for runtime validation
Tags + operationIds	`SensorsControllerMethods`, `ThresholdsControllerMethods`	Controllers `implements` these interfaces
Response schemas	`TemperatureCaptureResponse`, `ThresholdResponse`	Controller return types

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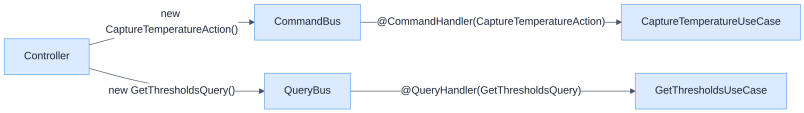
5. CQRS Mediator Pattern

Why CQRS?

Separating reads (queries) from writes (commands/actions) provides: - **Single Responsibility**: each use case handles exactly one operation - **Decoupling**: controllers don't know which use case handles their request - **Scalability**: queries and commands can be optimized independently

How It Works in NestJS



Conventions

COMPONENT	ROLE
`Action`	Write operation payload
`Query`	Read operation payload
`UseCase`	Business logic executor
`Validator`	Business logic validation
`CommandBus` / `QueryBus` (in Java we use our custome `BusinessEngine` with `primary` / `secondary` `ports` / `adapters`)	Mediator/router

Primary Ports, Why They Don't Exist Here

In Java, primary ports define the contract between controller and domain:

```
Controller → PrimaryPort (interface) → MediatorBusinessEngine → UseCase/Validators → Adapters (implements PrimaryPort)
```

In NestJS with CQRS, the **bus replaces the primary port**:

```
Controller → CommandBus.execute(Action) → UseCase (handler)
```

---

6. Hexagonal Architecture, Ports & Adapters

Controller (Api Core)

Application Layer = controller that implements the endpoints interfaces generated by the openAPI:

```
// src/application/api/api-core/controllers/threshold.controller.ts
@Get()
async thresholds(): Promise<ThresholdsResponse> {
  const result: Threshold = await this.queryBus.execute(new GetThresholdsQuery());
  return toThresholdResponse(result);
}
```

UseCase (Domain Core)

Business Logic = the core business logic of the application, contacts the infra using secondary ports, returns results to the api controllers:

```
// src/domain/domain-core/usecase/threshold/get-thresholds.usecase.ts
@QueryHandler(GetThresholdsQuery)
export class GetThresholdsUseCase implements IQueryHandler<GetThresholdsQuery, Threshold> {...}
```

Ports (Domain Contract)

Secondary Ports = interfaces that the domain needs (called by use cases, implemented by infrastructure):

```
// src/domain/domain-contract/ports/secondary/threshold.repository.port.ts
export interface ThresholdRepositoryPort {
  getCurrent(): Promise<Threshold | null>;
  update(coldMax: number, hotMin: number): Promise<Threshold>;
}
```

Adapters (Infrastructure)

Repository Adapters = implement the ports using TypeORM:

```
// src/infrastructure/src/persistence/adapters/threshold.repository.adapter.ts
@Injectable()
export class ThresholdRepositoryAdapter implements ThresholdRepositoryPort {
  constructor(@InjectRepository(ThresholdEntity) private readonly repo: Repository<ThresholdEntity>)
{}

  async getCurrent(): Promise<Threshold | null> {
    const entity = await this.repo.findOne({ where: {}, order: { updatedAt: 'DESC' } });
    return toDomain(entity);
  }
}
```

Injection Tokens

TypeScript interfaces disappear at runtime. NestJS cannot inject by interface type. We use **Symbols** as injection tokens:

Bridging compile-time interfaces with runtime implementations.

```
// src/shared/dinjection/tokens/injection-tokens.ts
export const THRESHOLD_REPOSITORY = Symbol('THRESHOLD_REPOSITORY');

// Infrastructure module, registers the implementation
{ provide: THRESHOLD_REPOSITORY, useClass: ThresholdRepositoryAdapter }

// UseCase, injects via token
constructor(@Inject(THRESHOLD_REPOSITORY) private readonly repo: ThresholdRepositoryPort) {}
```

Persistence Mappers

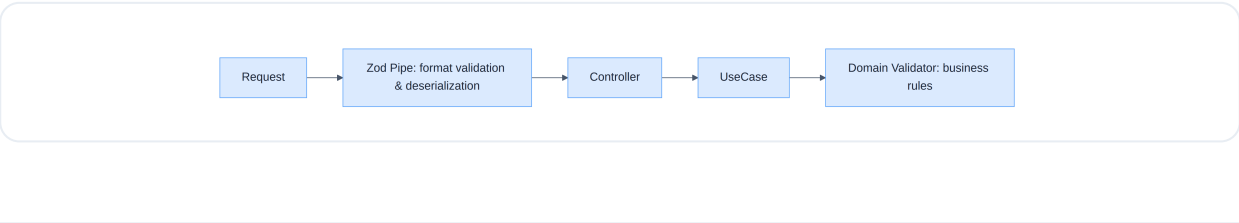
Mappers convert between infrastructure entities and domain models. They handle `null` (when no entity found):

```
export function toDomain(entity: ThresholdEntity | null): Threshold | null {
  if (!entity) return null;
  return {
    id: entity.id,
    coldMax: Number(entity.coldMax),
    hotMin: Number(entity.hotMin),
    updatedAt: entity.updatedAt
  };
}
```

---

7. Validation Strategy

Two Layers of Validation



LAYER	RESPONSIBILITY	EXAMPLE	HTTP STATUS
Zod (API)	Format, types, bounds from OpenAPI	`coldMax` must be a number between -50 and 60	400
Domain Validator	Business rules	`coldMax` must be less than `hotMin`	400 (ValidationException)

Zod Validation (Generated from OpenAPI)

The `ZodValidationPipe` uses schemas **generated automatically** from `openapi.yaml`:

Zero manual maintenance → change the YAML, regenerate, validation updates automatically:

```
# openapi.yaml
UpdateThresholdsRequest:
  properties:
    coldMax:
      type: number
      minimum: -50
      maximum: 60
```

Generates:

```
// zod.gen.ts (auto-generated, never edited manually)
export const zUpdateThresholdsRequest = z.object({
  coldMax: z.number().gte(-50).lte(60),
  hotMin: z.number().gte(-50).lte(60),
});
```

Used in controller:

```
@Put()
@UsePipes(new ZodValidationPipe(zUpdateThresholdsRequest))
async updateThresholds(@Body() body: UpdateThresholdsRequest): Promise<UpdateThresholdsResponse>
{ ... }
```

---

8. Error Handling

Exception Converters

Following the pattern from harvest-connect (`*ExceptionConverter`), each domain exception type has a dedicated converter:

Note: I used a centralized exception for all the domain exceptions just for simplicity, but in a real world case, each error should have a dedicated exception and exception converter to handle it, to make it easier for the client to understand the error and for us to find the bug and fix it.

EXCEPTION	CONVERTER	HTTP STATUS	MEANING
`ValidationException`	`ValidationExceptionConverter`	400	Business rule violated
`DomainException`	`DomainExceptionConverter`	422	Domain cannot process
`Error` (unhandled)	NestJS default	500	Unexpected

Registration order matters → `ValidationExceptionConverter` is registered first because `ValidationException extends DomainException`. NestJS uses the **most specific** filter that matches.

Error Response Shape

```
{
  "statusCode": 422,
  "code": "DomainException",
  "message": "No threshold configuration found",
  "timestamp": "2026-06-14T12:00:00.000Z",
  "path": "/api/v1/sensors/capture"
}
```

---

9. Testing Strategy

Architecture (inspired by Harvest-Connect backend (my current tasks in PlatformTeam))

TEST TYPE	LAYER	WHAT'S MOCKED	DATABASE	PURPOSE
Domain Unit	Domain	Secondary ports (`jest.fn()`)	❌	Test use case logic in isolation
Validator Unit	Domain	Nothing (pure function)	❌	Test business rules
Exception Converter	API	`ArgumentsHost` (mock)	❌	Test exception → HTTP mapping
API Error	API	`CommandBus` / `QueryBus` (mock)	❌	Test error response format
Integration	Full stack	Nothing	✅ Testcontainers	Full flow: HTTP → DB
Infrastructure	Infra	Nothing	✅ Testcontainers	Test adapter correctness

Test Naming Convention

Following harvest-connect pattern:

```
methodUnderTest_should_rxpectedBehavior_when_condition
```

Examples:

```
'getTemperatureHistory_should_return200WithArray_when_capturesExist'
```

Regions

Tests are organized with `//region` comments for collapsibility:

```
//region Success scenarios
it('execute_shouldReturnCapture_whenThresholdExists', ...);
//endregion

//region Error scenarios
it('execute_shouldThrowDomainException_whenNoThresholdFound', ...);
//endregion
```

Domain Tests Mocks

Following Java's use of `@InjectMock` (Mockito), we use `jest.fn()`:

```
let thresholdRepository: jest.Mocked<ThresholdRepositoryPort>;

beforeEach(() => {
  thresholdRepository = { getCurrent: jest.fn(), update: jest.fn() };
  usecase = new GetThresholdsUseCase(thresholdRepository);
});

it('execute_shouldThrowDomainException_whenNoThresholdFound', async () => {
  thresholdRepository.getCurrent.mockResolvedValue(null);
  await expect(usecase.execute()).rejects.toThrow(DomainException);
});
```

Integration Tests with Real Database (Testcontainers)

Integration tests boot the **full NestJS application** with a real PostgreSQL container:

```
beforeAll(async () => {
  container = await new GenericContainer('postgres:16-alpine')
    .withExposedPorts(5432)
    .withWaitStrategy(Wait.forLogMessage('database system is ready to accept connections', 2))
    .start();
  // ... wire real TypeORM + real adapters
});
```

No stubs, no mocks, the full flow from HTTP request to database and back.

---

10. Naming Conventions

> *Source: NestJS File and Folder Naming*

ELEMENT	CONVENTION	EXAMPLE
Files	`kebab-case.type.ts`	`capture-temperature.usecase.ts`
Classes	`PascalCase`	`CaptureTemperatureUseCase`
Interfaces	`PascalCase` + `Port` suffix	`ThresholdRepositoryPort`
Actions	`PascalCase` + `Action` suffix	`CaptureTemperatureAction`
Queries	`PascalCase` + `Query` suffix	`GetThresholdsQuery`
Mappers	pure functions, `camelCase`	`toThresholdResponse()`, `toDomain()`
Test files	`*.spec.ts` / `*.integration-spec.ts` / `*.api-spec.ts`	`domain-excetion.converter.spec.ts`

11. Conventional Commits

> *Source: Conventional Commits v1.0.0*

Format:

```
<type>(scope): JIRA-KEY - description
---
Types [ feat, fix, test, ci, docs, chore, refactor]
```

11. Commands Reference

Build & Run

Clean + generate OpenAPI types + compile TypeScript

```
npm run build
```

Build then run (`node dist/main`)

```
npm run start
```

Watch mode (hot reload)

```
npm run start:dev
```

Generate types/zod/nestjs from OpenAPI YAML

```
npm run generate
```

Remove `dist/` and `generated/`

```
npm run clean
```

Run linter pour checker

```
npm run lint
```

Fix code format with linter

```
npm run format
```

Test

Unit + API tests

```
npm run test:all
```

API tests (error + integration + exception)

```
npm run test:api
```

Domain unit tests only

```
npm run test:domain
```

Domain + exception converters tests

```
npm test
```

Integration tests only (needs Docker)

```
npm run test:integration
```

Infrastructure adapter tests (needs Docker)

```
npm run test:infra
```

Unit tests with coverage report inside root/converage folder

```
npm run test:cov
```

Docker

Start PostgreSQL only

```
docker compose up -d postgres
```

Start app + PostgreSQL

```
docker compose up
```

Stop all + remove volumes (reset DB)

```
docker compose down -v
```

---

12. Docker & Deployment

Dockerfile (Multi-stage)

```
FROM node:20-alpine AS builder
WORKDIR /app
# --> Builder : install deps + generate + compile

FROM node:20-alpine AS production
WORKDIR /app
COPY --from=builder /app/dist ./dist
COPY --from=builder /app/node_modules ./node_modules
COPY --from=builder /app/package.json ./
COPY --from=builder /app/src/application/api/api-contract/openapi.yaml ./src/application/api/api-
contract/openapi.yaml
# --> Production : minimal image
```

Key decisions: - `dumb-init` for proper signal handling (PID 1 problem) - `USER node` for security (non-root) - OpenAPI YAML copied for runtime Swagger UI serving

Docker Compose

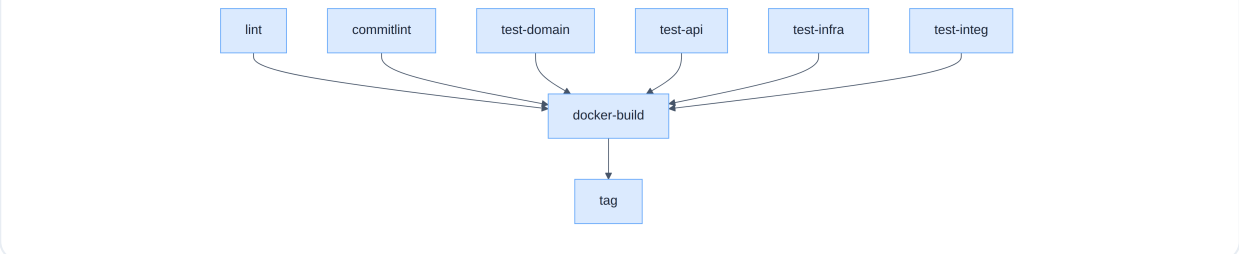
PostgreSQL init scripts are mounted as individual files (not subdirectories) because `docker-entrypoint-initdb.d/` only executes files at its root level:

```
volumes:
  - ./src/infrastructure/src/resources/db/migrations/001_initial_schema.sql:/docker-entrypoint-
initdb.d/01_schema.sql
  - ./src/infrastructure/src/resources/db/seeds/001_default_thresholds.sql:/docker-entrypoint-
initdb.d/02_seeds.sql
```

---

13. CI/CD Pipeline

GitHub Actions workflow (`.github/workflows/ci.yml`):



Appendix: Environment Configuration

> *Source: NestJS Environment Variables*

Configuration files in `config/` : - `.env` : development - `.env.test` : test - `.env.example` : template (committed)

Loaded via `ConfigModule.forRoot({ envFilePath: 'config/.env' })`.

---

14. Problems Encountered & Decisions

Problem 1: Runtime Enums from OpenAPI

Issue: `@hey-api/typescript` generates **type unions** (`'HOT' | 'COLD' | 'WARM'`), not runtime values. Tests couldn't use `Object.values(TemperatureState)` for assertions.

Solution: Added `enums: 'javascript'` to the plugin config. Now generates:

```
export const TemperatureState = { HOT: 'HOT', COLD: 'COLD', WARM: 'WARM' } as const;
export type TemperatureState = typeof TemperatureState[keyof typeof TemperatureState];
```

Both a runtime object AND a TypeScript type.

Problem 2: Validation Without Manual DTO Classes

Issue: NestJS `ValidationPipe` requires classes with `class-validator` decorators. This duplicates constraints already defined in OpenAPI.

Solution: Replaced `class-validator` with **Zod** (generated from OpenAPI via hey-api `zod` plugin) + a custom `ZodValidationPipe`. Zero manual maintenance, YAML is the single source.

Problem 3: Controller Interface Naming

Issue: The `nestjs` plugin generates `SensorsControllerMethods` , we wanted `SensorsControllerApi` .

Decision: Kept the default , it's hardcoded in the plugin and not configurable. Accepted as-is.

Problem 4: PostgreSQL docker-entrypoint-initdb.d Subdirectories

Issue: PostgreSQL init only executes `.sql` files **directly** in `docker-entrypoint-initdb.d/` , not in subdirectories.

Solution: Mount individual SQL files (not directories) with ordered prefixes (`01_schema.sql` , `02_seeds.sql`).

Problem 5: Testcontainers Wait Strategy

Issue: `Wait.forHealthCheck()` was unreliable , tests got `ECONNRESET` because PostgreSQL wasn't ready.

Solution: `Wait.forLogMessage('database system is ready to accept connections', 2)` , waits for PostgreSQL to print this message twice (once on initial start, once after recovery).

Problem 6: TypeScript Interfaces Disappear at Runtime

Issue: Can't use `@Inject(ThresholdRepositoryPort)` because interfaces don't exist at runtime in JavaScript.

Solution: Use `Symbol` injection tokens as the bridge between compile-time interfaces and runtime DI.

Problem 7: PostgreSQL Decimal Columns Return Strings

Issue: `decimal(5,2)` columns are returned as strings (`'25.50'` not `25.5`) by the `pg` driver.

Solution: Persistence mappers cast with `Number()` : `coldMax: Number(entity.coldMax)` .

Problem 8: Stubs vs Mocks in Domain Tests

Issue: Initially used in-memory stubs (mini-repositories with sorting logic). This violated the stub principle , stubs shouldn't contain logic.

Decision: Replaced all stubs with `jest.fn()` mocks for domain tests (like harvest-connect uses `@InjectMock` + Mockito). Stubs removed.

Problem 9: Integration Tests Without Real DB

Issue: Initial "integration" tests used in-memory stubs , not true integration tests.

Decision: Rewrote integration tests with testcontainers (real PostgreSQL). Full stack: HTTP → Controller → UseCase → Adapter → Database.

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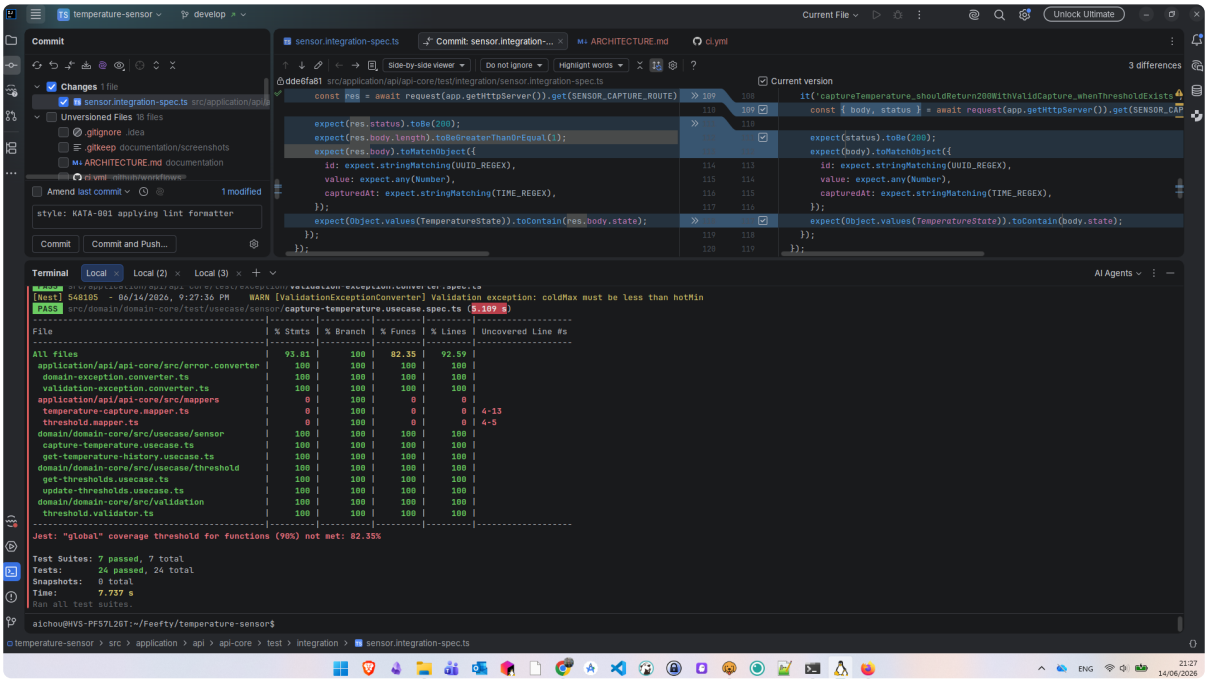
16. Screenshots

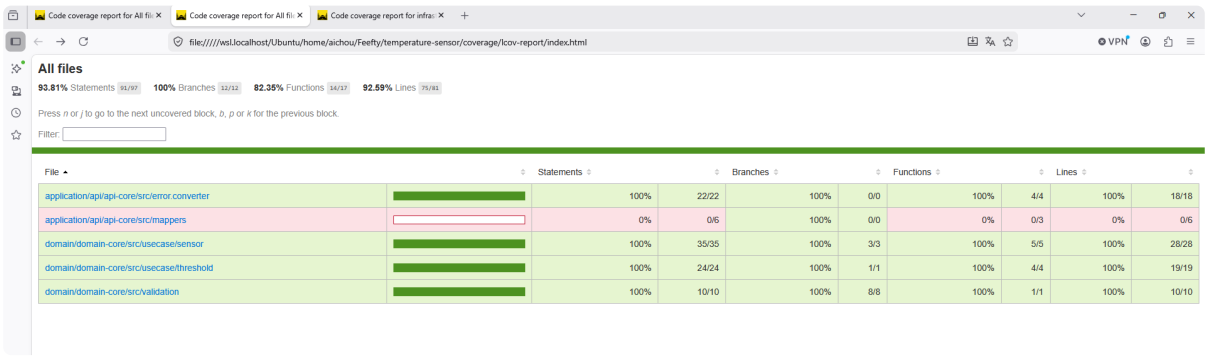
Coverage Report

`npm run test:cov`



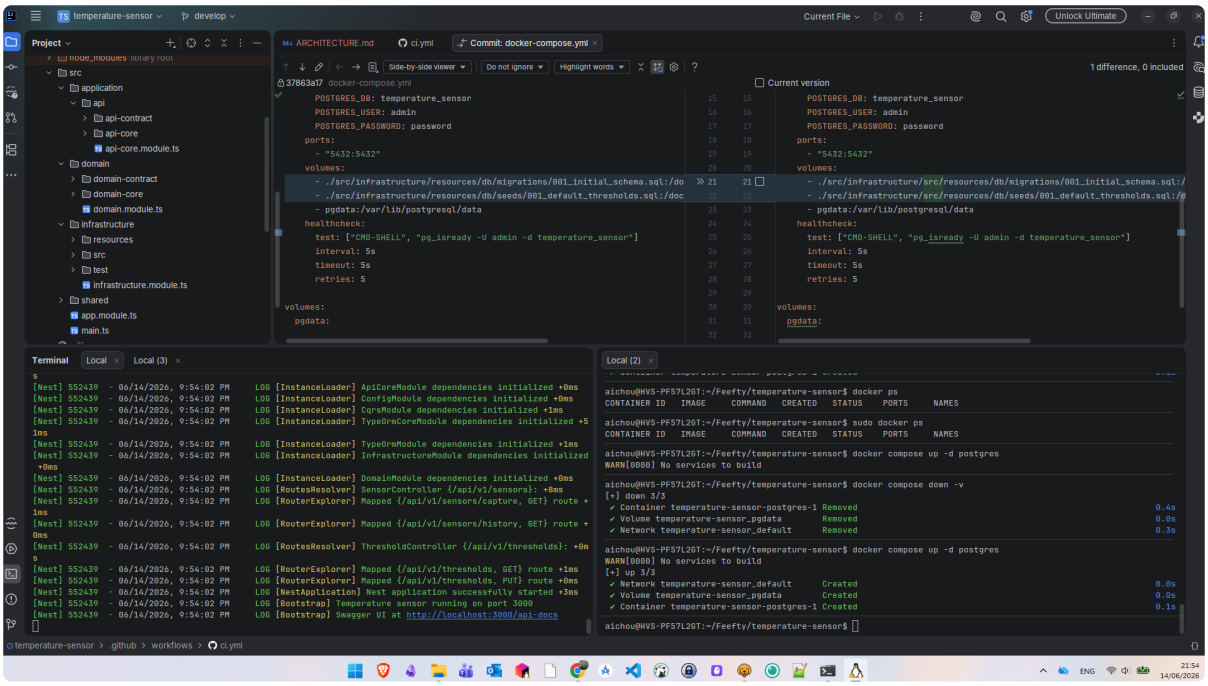
Coverage Summary

`coverage/lcov-report/index.html` browser



Coverage HTML Detail

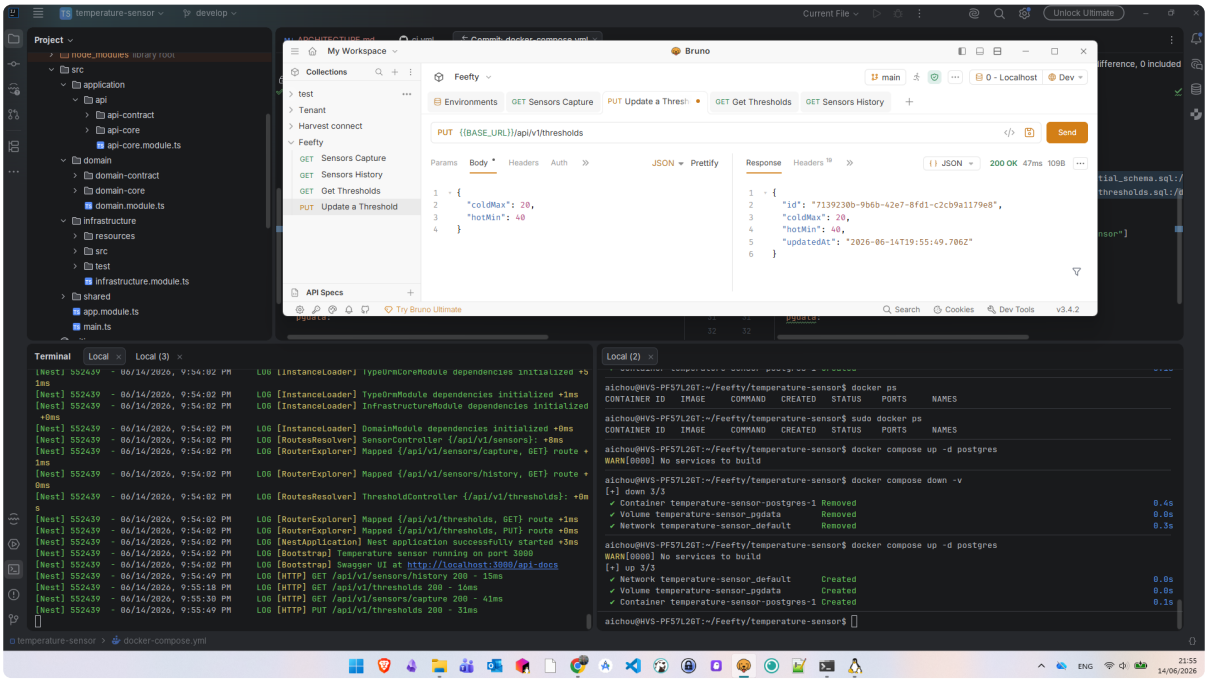
Application Running



Swagger UI

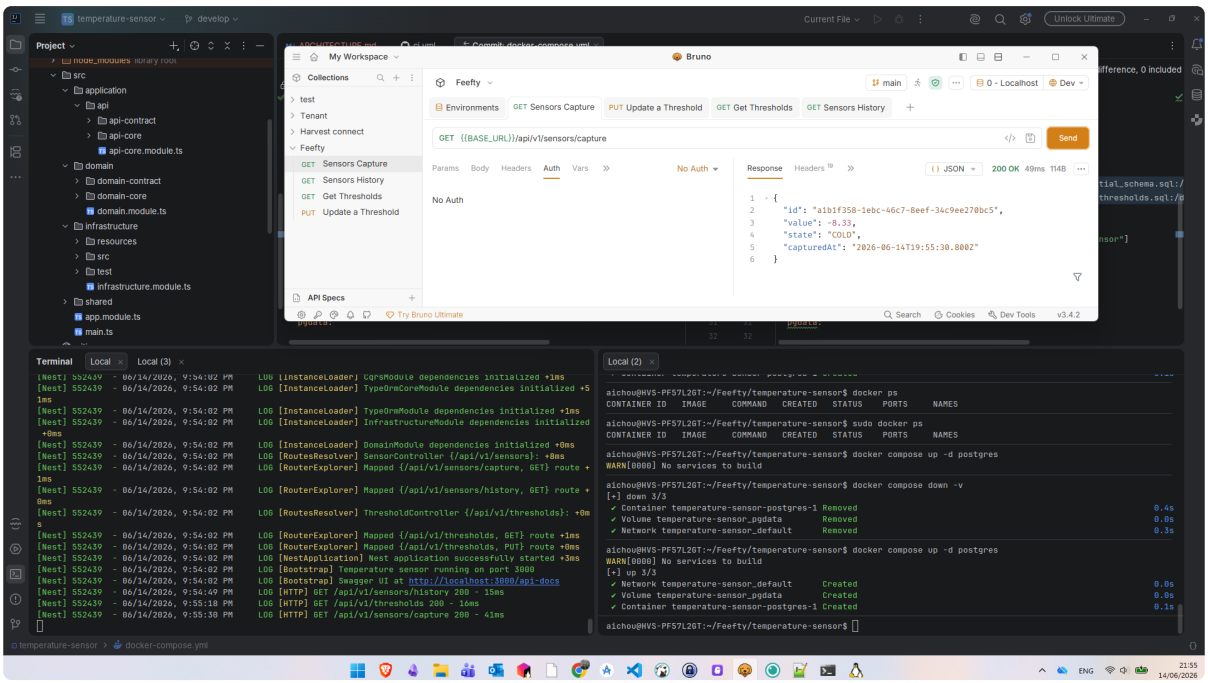
Testing with `Bruno` in local:

`PUT threshold`



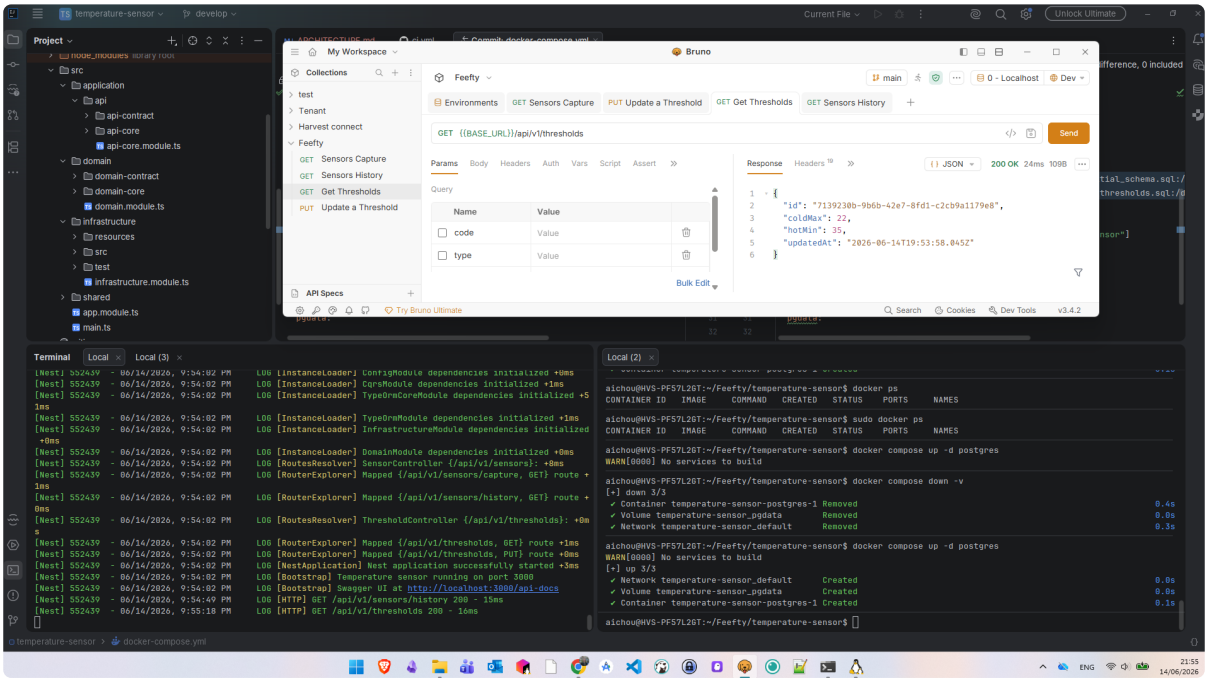
API Responses

`GET sensor capture`



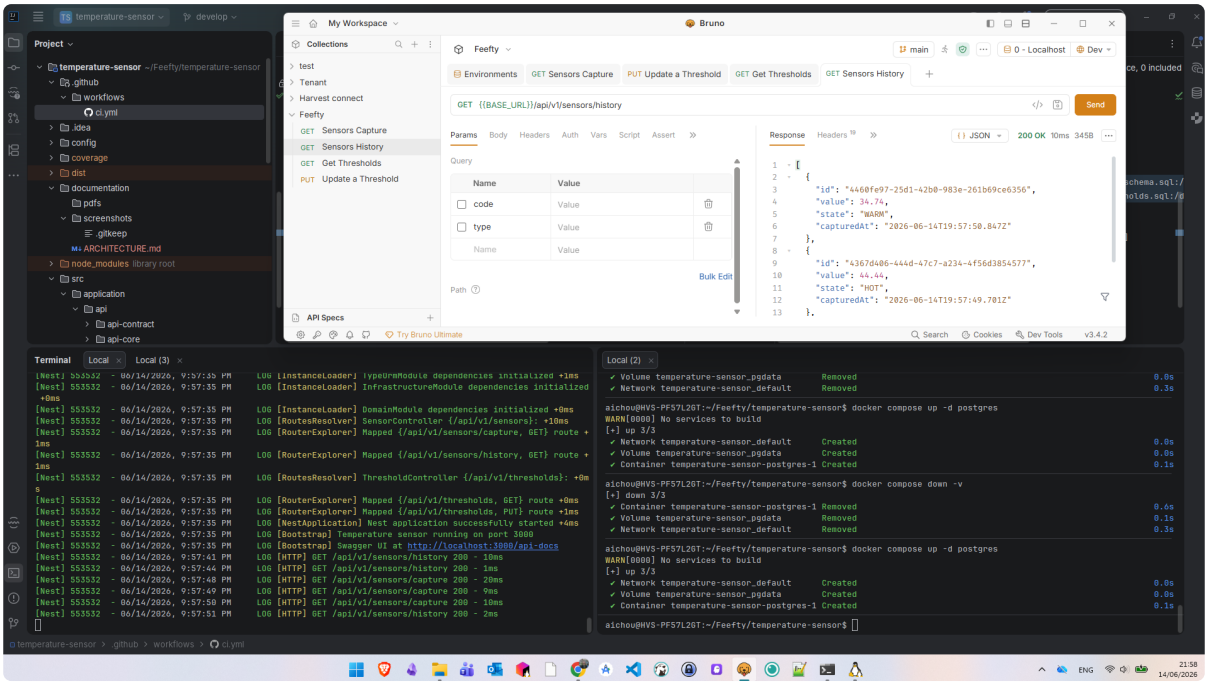
API Responses

`GET thresholds`



API Responses

`GET sensor history`



API Responses

Project created for Feefty from Harvest Groupe